E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23615229

RECEIVED 29.04.2026

ACCEPTED 02.05.2026

PUBLISHED 30-06-2026

Phytochemical Constituents of Amorphophallus paeoniifolius (Dennst.) Nicolson and Their Antioxidant Potential: A Comprehensive Review

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ABSTRACT:

Amorphophallus paeoniifolius (Dennst.) Nicolson, belonging to the family Araceae, is a widely distributed tuberous plant that occupies a prominent position in traditional medical systems across South Asia, Southeast Asia, and parts of East Africa. Commonly known as elephant foot yam, this plant has been integral to Ayurvedic, Siddha, and tribal pharmacopoeias for centuries, where it is employed in the management of conditions ranging from haemorrhoids and hepatic disorders to inflammatory diseases and metabolic disturbances. Despite its extensive ethnobotanical use, a systematic scientific understanding of the chemical diversity underpinning its therapeutic properties has only recently begun to emerge. This review consolidates published experimental data on the isolation and characterisation of phytochemical constituents — including alkaloids, flavonoids, phenolic acids, saponins, tannins, sterols, triterpenoids, polysaccharides, and glycosides — from different parts of the plant, with particular emphasis on the corm. The antioxidant potential of these constituents is critically evaluated through standard in vitro assays including DPPH, ABTS, FRAP, and phosphomolybdenum methods, alongside estimation of total phenolic and flavonoid content. The review further discusses the correlation between phytochemical composition and observed antioxidant capacity, addresses the influence of extraction solvents and methods on bioactive yield, and identifies gaps in the existing literature that warrant future investigation including in vivo validation, structure-activity relationships, and clinical translation.

Keywords: *Amorphophallus paeoniifolius, elephant foot yam, phytochemicals, antioxidant activity, DPPH, FRAP, phenolic content, flavonoids, glucomannan, Araceae.*

1. Introduction

The plant kingdom represents an extraordinarily rich repository of structurally diverse secondary metabolites, many of which have been exploited for medicinal purposes since antiquity. Among the taxonomically diverse group of tuber-bearing monocots, Amorphophallus paeoniifolius (Dennst.) Nicolson stands out as a species of exceptional nutritional and pharmacological significance. Belonging to the family Araceae and the order Alismatales, this perennial herbaceous plant produces a large underground corm — commonly called elephant foot yam owing to its size and rough, greyish-brown surface texture — that serves simultaneously as a staple food source and a versatile medicinal agent across rural communities in India, Bangladesh, Sri Lanka, Myanmar, Indonesia, the Philippines, and parts of tropical Africa [1, 2].

In Ayurvedic classical texts, the plant is referred to as "Soorana" or "Jimikand" and is listed among the drugs prescribed for arshas (haemorrhoids), gulma (abdominal tumour/bloating), pleeha (splenic disorders), and vibandha (constipation). The Charaka Samhita and Sushruta Samhita both mention its corm as possessing ushna (hot) potency, making it suitable for conditions attributed to vitiation of kapha and vata doshas [3]. Siddha practitioners in Tamil Nadu similarly prescribe the plant for skin ailments, inflammatory conditions, and respiratory disorders. Tribal communities in Chhattisgarh and Odisha deploy the raw plant material, often in paste or poultice form, for snakebite first aid, joint pain, and wound management, although the presence of calcium oxalate raphides necessitates heat processing before any internal use [4].

Beyond its traditional applications, Amorphophallus paeoniifolius has attracted growing scientific attention as a source of structurally interesting phytochemicals with demonstrable in vitro biological activity. The corm, petioles, and leaves have all been investigated using a variety of extraction solvents — ranging from water and ethanol to methanol, ethyl acetate, and n-hexane — yielding fractions enriched in different classes of secondary metabolites. Alkaloids such as betulinic acid and tricontane, flavonoids including quercetin and kaempferol, saponins based on the diosgenin skeleton, phenolic acids, tannins, and the structurally unique polysaccharide konjac glucomannan have all been identified and characterised using chromatographic and spectroscopic methods [5, 6]. Antioxidant evaluation, conducted primarily through DPPH and FRAP assays, has consistently demonstrated moderate to high free radical scavenging capacity that correlates with total phenolic and flavonoid content [7].

Despite this growing body of data, the published literature remains fragmented, with individual studies focusing on isolated phytochemical classes or single assay methods, and few attempts at integrating the totality of available evidence. This review aims to fill that gap by providing a structured, critical synthesis of the current state of knowledge on the phytochemistry and antioxidant evaluation of Amorphophallus paeoniifolius. The review additionally discusses methodological considerations, structure-activity relationships, and directions for future research including in vivo validation and clinical translation of the most promising bioactive constituents.

2. Botanical Description and Distribution

2.1 Taxonomy and Synonymy

Amorphophallus paeoniifolius (Dennst.) Nicolson belongs to the genus Amorphophallus within the family Araceae, tribe Thomsonieae. The genus name is derived from the Greek words "amorphos" (shapeless) and "phallos" (phallus), referring to the unusual shape of the spadix. The species epithet "paeoniifolius" indicates leaves resembling those of the peony. The plant was originally described by Dennstedt and later recombined by Nicolson in 1977. Accepted synonyms include Amorphophallus campanulatus (Roxb.) Blume ex Decne., Caladium paeoniifolium Dennst., Arum campanulatum Roxb., and Dracontium polyphyllum L. [2]. In India, it is known regionally as suran, jimikand, or oal (Hindi), oal (Bengali), kanda (Telugu), senaikizhangu (Tamil), and suvarna gadde (Kannada) [8].

2.2 Morphology

The plant produces a massive, depressed-globose corm measuring up to 30 cm in diameter and weighing between 1 and 8 kilograms at maturity, with a rough, fissured brown surface covered by adventitious roots. A single large compound leaf, typically 1 to 1.5 metres in height, emerges from the corm each growing season on a stout, spotted petiole. The leaf blade is tri-pinnately divided, with leaflets of varying sizes giving it a sprawling, umbrella-like appearance. Flowering precedes leaf emergence and takes place during the dry season; the inflorescence consists of a large spathe surrounding a central spadix that bears a characteristic foetid odour to attract pollinators. The fruit is an orange to red berry, each containing one or two seeds [9].

2.3 Habitat and Cultivation

Amorphophallus paeoniifolius is predominantly a plant of tropical and subtropical lowlands, thriving in humid, well-drained soils at altitudes below 800 metres. It is cultivated as a seasonal crop across the Indo-Gangetic plains, Chhattisgarh, Odisha, West Bengal, Maharashtra, and Andhra Pradesh in India, and is also found growing wild in forest margins and disturbed habitats. The plant exhibits a dormant phase during the dry season when the aboveground parts die back completely, with the corm persisting underground as the primary vegetative storage organ. Cultivation requires minimal inputs and the crop is highly tolerant of partial shade, making it a common component of agroforestry systems [10].

3. Phytochemical Constituents

3.1 Overview of Extraction Strategies

The chemical profile of Amorphophallus paeoniifolius varies considerably with the choice of extraction solvent, plant part used, and pre-processing treatment applied to the raw material. Polar solvents such as water, aqueous ethanol (50–80%), and methanol are most effective at extracting phenolic acids, flavonoids, glycosides, and water-soluble saponins. Non-polar solvents including n-hexane, petroleum ether, and chloroform selectively extract sterols, triterpenoids, fatty acids, and waxes. Ethyl acetate occupies an intermediate polarity and efficiently co-extracts flavonoid aglycones and moderately polar phenolics. Sequential exhaustive extraction — proceeding from non-polar to increasingly polar solvents — is therefore the preferred approach when a comprehensive phytochemical inventory is sought [11]. The corm, which is the primary pharmacologically active organ, must first be dried (sun-dried, shade-dried, or oven-dried at 50°C)

and powdered, since the high moisture and oxalate content of fresh material interfere with extraction efficiency.

3.2 Alkaloids

Alkaloids constitute one of the most pharmacologically active phytochemical classes in Amorphophallus paeoniifolius. Betulinic acid, a lupane-type pentacyclic triterpenoid alkaloid, has been isolated from the corm by column chromatography following petroleum ether extraction and characterised by infrared spectroscopy and nuclear magnetic resonance [12]. Betulinic acid has received considerable attention in oncology research owing to its selective pro-apoptotic activity against melanoma and several other carcinoma cell lines through activation of the intrinsic mitochondrial pathway. Tricontane, a long-chain alkane, has also been detected in non-polar fractions, alongside minor alkaloid constituents that have not yet been fully characterised. The nitrogen-containing heterocyclic compound amorphine has been reported by some investigators in leaf and petiole extracts, though its structural identity remains to be confirmed by high-resolution mass spectrometry [13].

3.3 Flavonoids

Flavonoids are among the most abundant polyphenolic constituents of Amorphophallus paeoniifolius and are held largely responsible for its antioxidant activity. Quercetin, a flavonol with well-established free radical scavenging, anti-inflammatory, and anti-carcinogenic properties, has been identified in both corm and leaf extracts by high-performance thin-layer chromatography (HPTLC) and confirmed by high-performance liquid chromatography coupled with diode array detection (HPLC-DAD) [14]. Kaempferol and its 3-O-glucoside (astragalin) have similarly been detected in ethanolic extracts. Rutin (quercetin-3-O-rutinoside) contributes significantly to the antioxidant profile of the leaf fraction, while luteolin and apigenin have been identified in smaller quantities. The flavonoid content of the corm increases with altitude of cultivation and declines markedly upon prolonged heat processing, suggesting that cooking methods influence the pharmacological potency of consumed preparations [15].

3.4 Phenolic Acids

Phenolic acids contribute substantially to the antioxidant capacity of corm extracts. Gallic acid, the simplest hydroxybenzoic acid, is consistently the most abundant phenolic acid in both aqueous and methanolic fractions, with concentrations of up to 4 mg per gram of dry weight reported in some studies [16]. Caffeic acid, ferulic acid, p-coumaric acid, and chlorogenic acid have been identified in the corm, petiole, and leaf extracts using HPLC-MS. Ferulic acid is particularly significant as it forms ester linkages with cell wall polysaccharides, contributing to structural integrity and also exhibiting UV-absorbing and free radical-quenching properties. Synergistic interactions between gallic acid and flavonoids such as quercetin have been proposed to explain the higher-than-additive antioxidant capacity of crude extracts relative to individual pure compounds [17].

3.5 Saponins

Saponins isolated from the corm of Amorphophallus paeoniifolius include diosgenin, a steroidal sapogenin that has attracted commercial interest as a precursor for the partial synthesis of steroid hormones and anti-inflammatory corticosteroids. Diosgenin is present at concentrations of 0.2 to 0.5% in dry corm

material and is obtained following acid hydrolysis of the corresponding steroidal glycosides [18]. Tigogenin and neotigogenin have also been reported as minor sapogenins. The intact glycoside fractions exhibit surface-active properties and demonstrate haemolytic activity at higher concentrations, which is a defining characteristic of saponins. Pharmacologically, corm saponins have been found to inhibit hepatic cholesterol synthesis, enhance biliary cholesterol excretion, and stimulate intestinal cholesterol clearance — effects consistent with the Ayurvedic prescription of the plant in jaundice and hepatic dysfunction [19].

3.6 Tannins

Tannins in Amorphophallus paeoniifolius include both condensed tannins (proanthocyanidins) and hydrolysable tannins, with condensed tannins predominating in the corm peel fraction. The protein-precipitating property of tannins underlies the astringent taste of raw corm preparations and contributes to their antimicrobial and antidiarrhoeic activities. The tannin content of the corm is substantially reduced upon boiling or autoclaving, which explains why processed elephant foot yam is well tolerated nutritionally [20]. Tannins also contribute indirectly to antioxidant activity by chelating metal ions that would otherwise catalyse Fenton-type generation of hydroxyl radicals.

3.7 Sterols and Triterpenoids

The non-polar fractions of Amorphophallus paeoniifolius corm are rich in phytosterols and pentacyclic triterpenoids. Beta-sitosterol, the predominant phytosterol, has been identified by gas chromatography-mass spectrometry (GC-MS) and characterised by its characteristic molecular ion at m/z 414 and prominent fragments at m/z 396 and 255. Stigmasterol and campesterol are co-occurring minor sterols. Triterpenoids including lupeol, oleanolic acid, and ursolic acid have been isolated by column chromatography over silica gel and characterised by NMR spectroscopy [21]. Lupeol is notable for its dual anti-inflammatory and anticancer properties, while oleanolic and ursolic acids inhibit topoisomerase II and exhibit cytotoxicity against multiple cancer cell lines at micromolar concentrations. All three triterpenoids also contribute to the overall antioxidant capacity of non-polar extracts.

3.8 Polysaccharides: Konjac Glucomannan

Among the macromolecular constituents of Amorphophallus paeoniifolius, the corm polysaccharide fraction deserves particular attention. The plant accumulates a high-molecular-weight water-soluble glucomannan in its corm parenchyma cells, structurally analogous to but distinct from the konjac glucomannan (KGM) characteristic of Amorphophallus konjac. This polymer consists of glucose and mannose residues in a molar ratio of approximately 1:1.6, linked by beta-1,4 glycosidic bonds with occasional beta-1,3 branches. Upon hydration, the polysaccharide forms a highly viscous gel that confers significant prebiotic activity by selectively stimulating growth of Bifidobacterium and Lactobacillus species in the colon [22]. The glucomannan fraction also demonstrates potent alpha-glucosidase inhibitory activity, explaining the antidiabetic effects observed in rodent feeding studies. At the antioxidant level, the polysaccharide itself is not a strong direct scavenger but may enhance overall antioxidant defence by modulating gut microbiota composition and thereby reducing systemic oxidative stress [23].

Table 1: Major Phytochemical Constituents of Amorphophallus paeoniifolius and Their Reported Biological Activities

S.No.	Class	Phytochemical Compound	Reported Biological Activity	Plant Part
1	Alkaloids	Amorphine, tricontane, betulinic acid	Antimicrobial, anti-inflammatory, anticancer	Corm
2	Flavonoids	Quercetin, kaempferol, rutin, luteolin	Antioxidant, anti-inflammatory, cardioprotective	Leaf, corm
3	Phenolic acids	Gallic acid, caffeic acid, ferulic acid, chlorogenic acid	Antioxidant, hepatoprotective, antimicrobial	Corm, leaf
4	Saponins	Diosgenin, tigogenin, neotigogenin	Hypocholesterolaemic, antidiabetic, anti-tumour	Corm
5	Tannins	Condensed tannins (proanthocyanidins), hydrolysable tannins	Antioxidant, astringent, antimicrobial	Corm, peel
6	Sterols	Beta-sitosterol, stigmasterol, campesterol	Anti-inflammatory, immunomodulatory	Corm
7	Triterpenoids	Lupeol, oleanolic acid, ursolic acid	Antioxidant, anti-inflammatory, hepatoprotective	Corm
8	Glycosides	Amorphophillin, steroidal glycosides	Antifungal, cardiotonic	Corm, petiole
9	Polysaccharides	Konjac glucomannan, amylase, mucilage	Prebiotic, antidiabetic, wound healing	Corm (tuber)

10	Oxalic acid crystals	Calcium oxalate raphides	Irritant (toxic if unprocessed)	All parts

4. Evaluation of Antioxidant Activity

4.1 Rationale for Antioxidant Assessment

Oxidative stress — defined as the imbalance between the generation of reactive oxygen species (ROS) and the capacity of cellular antioxidant defence mechanisms to neutralise them — is now recognised as a central pathophysiological driver in conditions including atherosclerosis, type 2 diabetes mellitus, neurodegeneration, cancer, and accelerated biological ageing [24]. Plants synthesise a diverse array of antioxidant secondary metabolites — phenolics, flavonoids, ascorbate, tocopherols — as part of their own ROS management systems, and dietary or medicinal consumption of these compounds has been associated with reduced disease incidence in epidemiological studies. The systematic evaluation of plant extracts for antioxidant capacity using standardised in vitro assays therefore provides a pharmacologically meaningful proxy for potential in vivo benefit, while the correlation of antioxidant activity with total phenolic and flavonoid content offers mechanistic insight into which compound classes drive the observed effects [25].

4.2 DPPH Free Radical Scavenging

The DPPH (1,1-diphenyl-2-picrylhydrazyl) assay is the most widely applied method for screening plant antioxidant activity owing to its simplicity, reproducibility, and the stability of the DPPH radical in methanol solution. The stable purple radical is reduced to yellow DPPH-H upon receiving a hydrogen atom or electron from an antioxidant compound, with the percentage inhibition quantified spectrophotometrically at 517 nm [26]. Ethanolic and methanolic extracts of the Amorphophallus paeoniifolius corm consistently demonstrate moderate to high DPPH scavenging activity. IC50 values in the range of 45 to 120 µg/mL have been reported across different studies, with the variation attributable to differences in geographical origin of the plant material, extraction conditions, and seasonal harvesting time. Quercetin and gallic acid, when present in high concentrations in the extract, have been identified as the principal contributors to DPPH scavenging capacity in phenolic-rich fractions [27]. Aqueous extracts typically display lower activity than ethanolic extracts for this assay, consistent with the greater solubility of flavonoid aglycones in ethanol compared to water.

4.3 ABTS Radical Cation Decolourisation

The ABTS (2,2'-azino-bis(3-ethylbenzothiazoline-6-sulphonic acid)) assay, also expressed as Trolox equivalent antioxidant capacity (TEAC), offers a complementary perspective by assessing the ability of antioxidants to quench both hydrophilic and lipophilic radicals. The blue-green ABTS radical cation generated by chemical or enzymatic oxidation is decolourised by antioxidants, and activity is referenced to Trolox [28]. Methanolic corm extracts of Amorphophallus paeoniifolius have demonstrated TEAC values comparable to those of ascorbic acid at equivalent concentrations in several studies, indicating substantial hydrogen-donating

or electron-transferring capacity. The ABTS assay tends to yield higher activity values than DPPH for the same extract because it can detect antioxidants across a wider polarity range.

4.4 Ferric Reducing Antioxidant Power (FRAP)

The FRAP assay measures the capacity of an extract to reduce ferric tripyridyltriazine (Fe3+-TPTZ) to its ferrous form (Fe2+-TPTZ), which develops an intense blue colour measurable at 593 nm [29]. This assay is particularly sensitive to phenolic compounds capable of single electron transfer. Aqueous and ethanolic leaf extracts of Amorphophallus paeoniifolius show high FRAP values at a concentration of 1 mg/mL, often exceeding those of n-hexane and ethyl acetate fractions. This finding is consistent with the higher phenolic acid content of polar extracts, since ferulic acid, caffeic acid, and gallic acid are all potent single-electron donors. A strong positive linear correlation ($r > 0.90$) between FRAP values and total phenolic content has been established in multiple studies, reinforcing the key role of phenolic acids in the antioxidant capacity of this plant [30].

4.5 Total Phenolic Content and Total Flavonoid Content

Quantification of total phenolic content (TPC) by the Folin-Ciocalteu method and total flavonoid content (TFC) by the aluminium chloride colorimetric method provides the chemical basis for interpreting antioxidant data. TPC values in Amorphophallus paeoniifolius corm extracts range from 23 to 98 mg gallic acid equivalents (GAE) per gram of dry weight across the literature, with ethanolic extracts consistently outperforming aqueous extracts [31]. TFC values expressed as quercetin equivalents (QE) range from 15 to 67 mg QE/g. The discrepancy between studies is not surprising given the variability in plant ecotype, cultivation conditions, post-harvest handling, and drying methodology. Regression analysis across multiple datasets indicates that TPC explains approximately 70 to 85 percent of the variance in DPPH IC50 values, identifying phenolics as the dominant antioxidant determinants [32].

4.6 Influence of Extraction Method on Antioxidant Yield

The selection of extraction methodology profoundly influences both the quantitative yield of bioactive compounds and the qualitative profile of the resulting extract. Comparative studies have established that maceration in 70% ethanol for 72 hours at room temperature provides a consistently higher phenolic yield than hot aqueous extraction, likely because extended contact time at ambient temperature minimises thermal degradation of heat-sensitive flavonoids and phenolic acids [33]. Ultrasonication-assisted extraction (UAE) significantly enhances extraction efficiency by disrupting cell walls through cavitation, typically achieving 20 to 40 percent higher TPC values relative to conventional maceration within a fraction of the extraction time. Microwave-assisted extraction (MAE) similarly improves yield but requires careful temperature control to prevent degradation of thermolabile compounds. Supercritical CO2 extraction, while excellent for non-polar terpenoid and sterol fractions, is poorly suited to phenolic compounds due to their low solubility in supercritical CO2 at typical operating pressures.

Table 2: In Vitro Antioxidant Assays Applied to Amorphophallus paeoniifolius Extracts

S.No.	Assay Method	Principle	Extract Used	Reported IC50 / Activity
1	DPPH (1,1-diphenyl-2-picrylhydrazyl) radical scavenging	Hydrogen atom or electron donation to DPPH radical; decolourisation measured at 517 nm	Ethanolic corm extract	45–120 µg/mL
2	ABTS radical cation (Trolox equivalent antioxidant capacity)	Oxidation of ABTS to green radical cation; inhibition by antioxidant measured at 734 nm	Methanolic corm extract	Comparable to ascorbic acid
3	Ferric reducing antioxidant power (FRAP)	Reduction of Fe3+ to Fe2+ in the presence of antioxidants; absorbance at 593 nm	Aqueous and ethanolic leaf extract	High reducing power at 1 mg/mL
4	Phosphomolybdenum total antioxidant capacity	Reduction of Mo(VI) to Mo(V); green phosphomolybdate complex at 695 nm	Ethanol corm extract	Moderate to high TAC
5	Hydrogen peroxide (H2O2) scavenging	Absorption of H2O2 in UV at 230 nm; % inhibition calculated	Aqueous corm extract	60–75% at 200 µg/mL

6	Hydroxyl radical scavenging (Fenton reaction)	Deoxyribose degradation assay; absorbance at 532 nm	Methanolic extract	Moderate scavenging activity
7	Total phenolic content (TPC) — Folin-Ciocalteu	Reduction of Folin reagent by phenols; blue colour at 765 nm; expressed as GAE	All solvent extracts	23–98 mg GAE/g dry weight
8	Total flavonoid content (TFC) — AlCl3 method	AlCl3 chelation with flavonoids; yellow colour at 415 nm; expressed as QE	Ethanolic extract	15–67 mg QE/g dry weight

5. Traditional Uses and Pharmacological Validation

The convergence of ethnobotanical practice and scientific investigation represents one of the most productive paradigms in natural product research. Amorphophallus paeoniifolius provides a compelling example of this convergence, with many of its traditionally reported uses now supported by experimental evidence at the molecular and cellular level. Table 3 summarises the principal traditional applications of the plant alongside the currently available scientific validation for each use. Hepatoprotective activity, one of the most widely cited traditional properties, has been demonstrated in carbon tetrachloride-induced hepatotoxicity models in rats, where corm extract administration significantly reduced serum transaminase levels and hepatic lipid peroxidation [34]. Saponins are believed to contribute to this activity by inhibiting cytochrome P450 2E1-mediated oxidative bioactivation of hepatotoxicants. Hypoglycaemic activity has been demonstrated in streptozotocin-induced diabetic rats using aqueous corm extracts, with a mechanism attributed to alpha-glucosidase inhibition by saponins and glucomannan-mediated slowing of glucose absorption [35]. Anti-inflammatory activity has been confirmed in carrageenan-induced paw oedema models in rats, with the corm extract comparing favourably with standard indomethacin at equivalent doses.

Table 3: Traditional Medicinal Uses of Amorphophallus paeoniifolius and Scientific Validation

S.No.	Traditional System	Reported Use	Scientific Evidence

1	Ayurveda (Soorana / Jimikand)	Haemorrhoids (piles), splenomegaly, digestive disorders, liver complaints	Corm extracts showed hepatoprotective activity; saponins reduce inflammation in anorectal tissues
2	Siddha medicine	Skin diseases, wound healing, itching	Antimicrobial and wound-healing activity confirmed in ethanolic extract; tannins contribute astringency
3	Tribal medicine (Central India)	Snake bite, scorpion sting, pain relief	Alkaloids including betulinic acid show anti-venom and analgesic properties in animal models
4	Folklore (Chhattisgarh, Odisha)	Bronchitis, cough, asthma, joint pain	Anti-inflammatory effects of sterols and flavonoids reduce prostaglandin synthesis; bronchospasmolytic activity noted
5	Chinese traditional medicine	Obesity, hyperlipidaemia, constipation	Konjac glucomannan increases satiety, reduces LDL cholesterol, and improves bowel transit time
6	Unani medicine	Kidney stones, urinary complaints	Diuretic and antilithiatic activity of aqueous extracts demonstrated in rat models

7	General folk use (South Asia)	Diabetes management, blood sugar control	Hypoglycaemic effects linked to glucomannan fibre and saponins; inhibition of alpha-glucosidase confirmed
8	Tribal use (Northeastern India)	Cancer prevention, anti-tumour	Lupeol, betulinic acid, and ursolic acid induce apoptosis in cancer cell lines in vitro

6. Structure-Activity Relationships of Key Antioxidant Constituents

Understanding the molecular basis of antioxidant activity in the phytochemicals of Amorphophallus paeoniifolius requires examination of how specific structural features govern free radical scavenging capacity. Among flavonoids, the presence of an ortho-dihydroxy (catechol) moiety on the B-ring — as in quercetin and luteolin — confers superior antioxidant capacity relative to flavonoids with a single hydroxyl group on the B-ring, such as kaempferol and apigenin. This is because the catechol B-ring can stabilise the flavonoxyl radical through resonance delocalisation and hydrogen bonding between the adjacent hydroxyl groups [36]. The 3-hydroxyl group on the C-ring and the 4-oxo group in conjugation with the 2,3-double bond provide additional radical stabilisation in flavonols. Glycosylation at the 3-position, as in quercetin-3-O-rutinoside (rutin), moderately reduces antioxidant activity by disrupting the planarity of the molecule and introducing steric effects, though rutin remains a significant antioxidant contributor owing to its high aqueous solubility and abundance.

Among phenolic acids, the antioxidant hierarchy is determined by the number and position of hydroxyl substituents on the aromatic ring, as well as the nature of the side chain. Gallic acid, with three hydroxyl groups at positions 3, 4, and 5, is a superior scavenger relative to caffeic acid (two hydroxyl groups on the catechol ring) and ferulic acid (one hydroxyl plus one methoxy group). The trans-configuration of the side chain double bond in caffeic and ferulic acids enhances conjugation with the aromatic ring, contributing to radical stabilisation through the vinyl phenol resonance structure [37]. Chlorogenic acid, the ester of caffeic acid with quinic acid, exhibits high antioxidant potency and contributes to the overall activity of polar corm extracts.

Triterpenoids including ursolic acid and oleanolic acid are moderate antioxidants in isolation but may exert indirect antioxidant effects in vivo by activating the Nrf2-Keap1 signalling pathway, upregulating the expression of endogenous antioxidant enzymes including haeme oxygenase-1 (HO-1), glutathione S-transferase, and NAD(P)H:quinone oxidoreductase 1 (NQO1) [38]. This mechanism of indirect antioxidation, while not captured by in vitro assays, may be of greater biological significance than direct radical scavenging in the context of chronic disease prevention.

7. Toxicological Considerations and Safety Profile

The safety of Amorphophallus paeoniifolius preparations is contingent upon adequate pre-processing of the raw plant material. The corm contains abundant calcium oxalate raphide crystals that cause intense oral and oropharyngeal irritation upon contact with mucous membranes, a property mediated by mechanical tissue penetration combined with possible enzymatic co-factors rather than oxalic acid toxicity per se [39]. Thorough cooking — whether by boiling, autoclaving, or fermentation — disrupts the raphide crystal structure and inactivates any associated irritant factors, rendering the cooked corm safe for consumption. Processed corm preparations used in pharmacological studies have consistently shown no acute toxicity in rodent models at doses up to 2000 mg/kg body weight administered orally, placing them in Globally Harmonized System (GHS) Toxicity Category 5 or unclassified [40].

Repeated dose oral toxicity studies have similarly reported no significant alterations in haematological, biochemical, or histopathological parameters at therapeutic doses in rodents. The glucomannan polysaccharide is recognised as a generally safe food ingredient in the European Union and has received GRAS (Generally Recognised As Safe) status from the United States Food and Drug Administration. One safety concern unique to hypoglycaemic individuals is the potential for additive blood glucose lowering when corm preparations are consumed concurrently with antidiabetic medications, warranting appropriate dose monitoring in this population.

8. Discussion

The accumulated body of phytochemical and antioxidant research on Amorphophallus paeoniifolius paints a coherent picture of a plant whose long-standing medicinal reputation is well grounded in chemical reality. The corm, as the primary pharmacologically active organ, harbours a remarkably diverse array of bioactive secondary metabolites spanning multiple structural classes. The dominance of polyphenolics — particularly flavonoids and phenolic acids — as determinants of antioxidant activity is consistent with the broad pattern observed across the plant kingdom, where these compounds serve dual roles as UV screens, pathogen deterrents, and ROS neutralisers [41]. However, several features of this particular species merit special comment.

First, the coexistence of a nutritionally valuable polysaccharide (glucomannan) with potent small-molecule antioxidants in the same organ is pharmacologically synergistic. The glucomannan fraction can modulate gut microbiota in ways that reduce systemic endotoxin exposure and consequent activation of the NLRP3 inflammasome — an increasingly recognised driver of oxidative tissue damage. Thus, consuming processed Amorphophallus paeoniifolius may confer antioxidant benefit not only through direct radical scavenging but also through microbiome-mediated reduction of endogenous ROS generation, an effect that no in vitro assay can capture [42].

Second, the correlation between phytochemical content and antioxidant activity, while statistically robust, should be interpreted with the caveat that in vitro antioxidant assays measure only thermodynamic and kinetic capacity under artificial conditions. The bioaccessibility of phenolic compounds after gastrointestinal transit, the degree of biotransformation by colonic microbiota to yield active metabolites such as urolithins from ellagitannins, and the cellular uptake and retention of individual flavonoids are all variables that

profoundly influence in vivo antioxidant efficacy but are invisible to DPPH and FRAP measurements [43]. Future research should incorporate cell-free bioaccessibility models, Caco-2 monolayer permeability studies, and ultimately metabolite profiling in plasma following oral dosing to bridge the gap between in vitro data and physiological relevance.

Third, the reported IC50 values for DPPH scavenging show a threefold variation across studies (45 to 120 µg/mL), which is too wide a range to draw firm conclusions about dose-activity relationships for any particular preparation. Standardisation of extraction and testing protocols is therefore a priority. The adoption of internationally harmonised methodologies — such as those described by Folin for TPC and by Brand-Williams for DPPH — alongside use of commercially certified reference standards and reporting of data as absolute IC50 with 95% confidence intervals, would substantially improve the comparability of results across laboratories.

9. Future Research Directions

Several promising research avenues remain to be explored to translate the extensive in vitro database on Amorphophallus paeoniifolius into clinically actionable knowledge. In vivo animal model studies using well-defined oxidative stress induction models — such as tert-butyl hydroperoxide-induced hepatic oxidative stress in mice, or high-fat diet-induced metabolic syndrome — would provide the first indication of whether the in vitro antioxidant capacity of corm extracts translates into measurable reductions in tissue lipid peroxidation markers (malondialdehyde), protein carbonyl content, and activity of endogenous antioxidant enzymes (superoxide dismutase, catalase, glutathione peroxidase) [44].

Isolation and structural characterisation of glucoside-linked flavonoids and novel phenolic glycosides using HPLC-HR-MS (high-resolution mass spectrometry) combined with multidimensional NMR would complete the chemical inventory that currently lacks detail on minor constituents. Bioactivity-guided fractionation of the most potent antioxidant fractions, combined with molecular docking against known antioxidant targets such as Nrf2 and KEAP1, would provide mechanistic insights without requiring immediate animal studies. Network pharmacology approaches — constructing target-compound interaction networks from publicly available databases — can rapidly identify which phytochemical constituents of Amorphophallus paeoniifolius are most likely to activate antioxidant gene expression programmes.

From a translational perspective, the formulation of standardised corm extract as a nutraceutical product — in tablet, capsule, or functional food form — with guaranteed minimum content of total phenolics (expressed as GAE/serving) and validated antioxidant activity would represent a meaningful step towards commercial development. Clinical pilot trials in populations with elevated oxidative stress burden — such as type 2 diabetics, patients with non-alcoholic fatty liver disease, or elderly subjects — would generate the human efficacy and safety data necessary for regulatory consideration.

10. Conclusion

Amorphophallus paeoniifolius (Dennst.) Nicolson is a botanically unique and pharmacologically rich species whose secondary metabolite profile encompasses a wide range of biologically relevant compound classes. The corm, as the principal medicinal organ, accumulates flavonoids (particularly quercetin and kaempferol), phenolic acids (gallic, caffeic, ferulic acids), saponins (diosgenin-based), tannins, sterols (beta-sitosterol), triterpenoids (lupeol, ursolic acid, oleanolic acid), alkaloids (betulinic acid), and the structural polysaccharide konjac glucomannan. In vitro antioxidant evaluation consistently reveals moderate to high activity in DPPH, ABTS, and FRAP assays, with total phenolic content as the dominant predictor of scavenging capacity. These chemical findings provide a sound molecular basis for the plant's extensive use in Ayurvedic, Siddha, and tribal medicine for conditions now understood to be driven in part by oxidative stress, including hepatic inflammation, metabolic dysfunction, and carcinogenesis.

To advance the field, future investigations must move beyond in vitro screening towards bioaccessibility studies, in vivo mechanistic models, and ultimately structured clinical evaluations. Standardisation of extraction and analytical protocols is equally necessary to improve cross-study comparability. With this research infrastructure in place, Amorphophallus paeoniifolius holds considerable promise as a source of clinically validated antioxidant nutraceuticals and as a scaffold for drug discovery targeting oxidative stress-mediated pathologies.

Acknowledgements

The authors sincerely thank the Faculty of Pharmacy, Kalinga University, Naya Raipur, Chhattisgarh, for providing the academic resources and institutional support that facilitated this work. Gratitude is also extended to the library staff for assistance with database access.

Conflict of Interest

The authors declare no conflicts of interest related to this review.

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